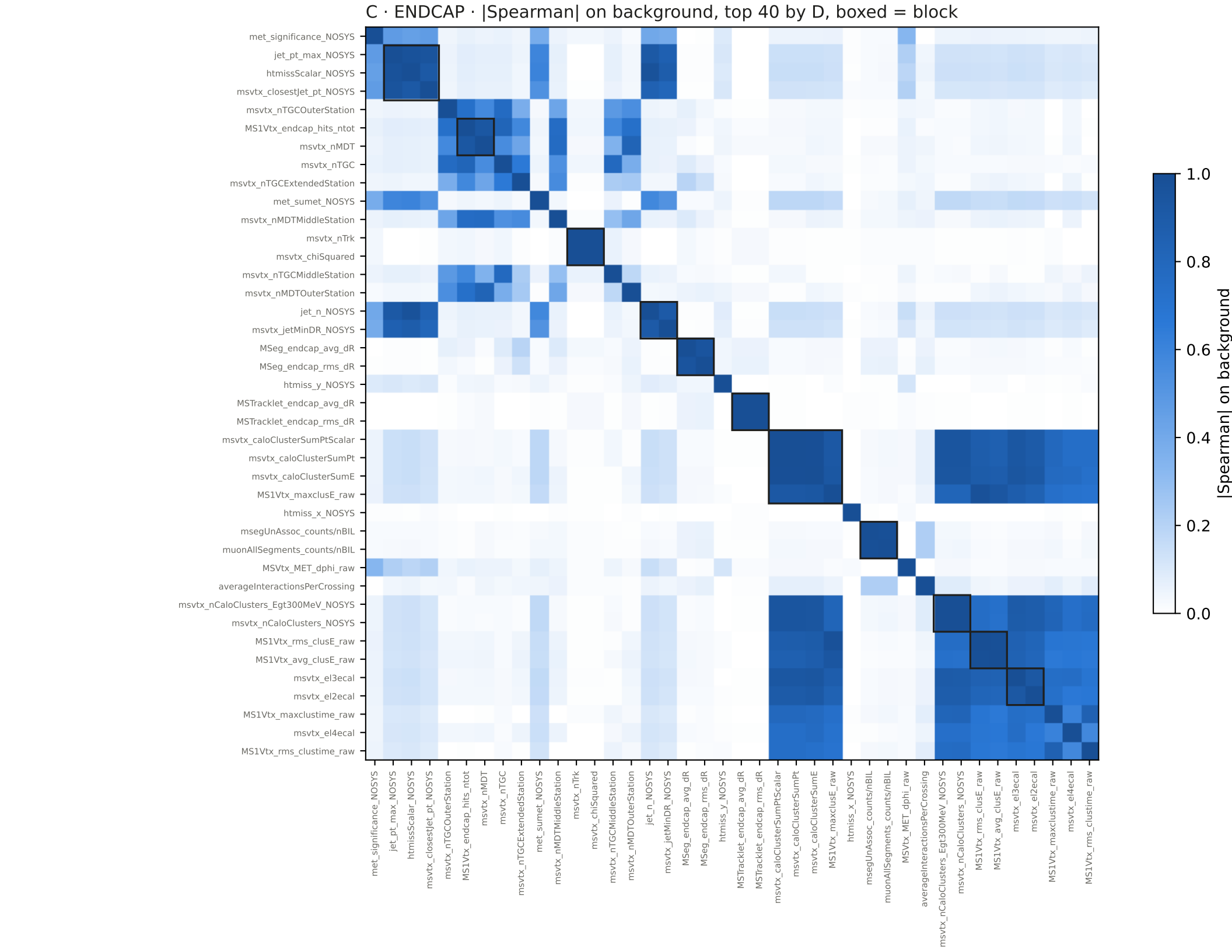

C · ENDCAP · |Spearman| on background, top 40 by D, boxed = block
met_significance_NOSYS
jet_pt_max_NOSYS
htmissScalar_NOSYS
msvtx_closestJet_pt_NOSYS
msvtx_nTGCOuterStation
MS1Vtx_endcap_hits_ntot
msvtx_nMDT
msvtx_nTGC
msvtx_nTGCExtendedStation
met_sumet_NOSYS
msvtx_nMDTMiddleStation
msvtx_nTrk
msvtx_chiSquared
msvtx_nTGCMiddleStation
msvtx_nMDTOuterStation
jet_n_NOSYS
msvtx_jetMinDR_NOSYS
MSeg_endcap_avg_dR
MSeg_endcap_rms_dR
htmiss_y_NOSYS
MSTracklet_endcap_avg_dR
MSTracklet_endcap_rms_dR
msvtx_caloClusterSumPtScalar
msvtx_caloClusterSumPt
msvtx_caloClusterSumE
MS1Vtx_maxclusE_raw
htmiss_x_NOSYS
msegUnAssoc_counts/nBIL
muonAllSegments_counts/nBIL
MSVtx_MET_dphi_raw
averageInteractionsPerCrossing
msvtx_nCaloClusters_Egt300MeV_NOSYS
msvtx_nCaloClusters_NOSYS
MS1Vtx_rms_clusE_raw
MS1Vtx_avg_clusE_raw
msvtx_el3ecal
msvtx_el2ecal
MS1Vtx_maxclustime_raw
msvtx_el4ecal
MS1Vtx_rms_clustime_raw
|Spearman| on background
1.0
0.8
0.6
0.4
0.2
0.0

C · ENDCAP · redundancy blocks

```
71 blocks; a feature joins a block only if |rho| >= 0.9 with every member;
the first member (strongest separator) is the representative.

block   2  rep jet_pt_max_NOSYS  (D 0.854)
            htmissScalar_NOSYS                      D 0.789   rho to rep +0.972
            msvtx_closestJet_pt_NOSYS               D 0.687   rho to rep +0.956
block   4  rep MS1Vtx_endcap_hits_ntot  (D 0.718)
            msvtx_nMDT                              D 0.672   rho to rep +0.926
block   9  rep msvtx_nTrk  (D 0.606)
            msvtx_chiSquared                        D 0.486   rho to rep +1.000
block  12  rep jet_n_NOSYS  (D 0.511)
            msvtx_jetMinDR_NOSYS                    D 0.493   rho to rep -0.905
block  13  rep MSeg_endcap_avg_dR  (D 0.462)
            MSeg_endcap_rms_dR                      D 0.384   rho to rep +0.952
block  15  rep MSTracklet_endcap_avg_dR  (D 0.399)
            MSTracklet_endcap_rms_dR                D 0.399   rho to rep +1.000
block  16  rep msvtx_caloClusterSumPtScalar  (D 0.393)
            msvtx_caloClusterSumPt                  D 0.393   rho to rep +1.000
            msvtx_caloClusterSumE                   D 0.383   rho to rep +0.989
            MS1Vtx_maxclusE_raw                     D 0.374   rho to rep +0.920
block  18  rep msegUnAssoc_counts/nBIL  (D 0.378)
            muonAllSegments_counts/nBIL             D 0.357   rho to rep +0.986
block  21  rep msvtx_nCaloClusters_Egt300MeV_NOSYS  (D 0.366)
            msvtx_nCaloClusters_NOSYS               D 0.365   rho to rep +0.998
block  22  rep MS1Vtx_rms_clusE_raw  (D 0.365)
            MS1Vtx_avg_clusE_raw                    D 0.361   rho to rep +0.987
block  23  rep msvtx_el3ecal  (D 0.364)
            msvtx_el2ecal                           D 0.343   rho to rep +0.923
block  35  rep nBOL_NOSYS  (D 0.271)
            muonAllSegments_counts/nBOL             D 0.253   rho to rep +0.988
block  43  rep muonSegments_nSeg  (D 0.201)
            msegUnAssoc_nSeg                        D 0.187   rho to rep +0.962
block  45  rep msegUnAssoc_counts/nEML  (D 0.187)
            muonAllSegments_counts/nEML             D 0.179   rho to rep +0.969
block  46  rep msegNCB_counts/nEIL  (D 0.151)
            muonAllSegments_counts/nEIL             D 0.119   rho to rep +0.918
block  47  rep msvtx_trackSoftSumNTracks  (D 0.140)
            msvtx_trackSoftSumPt_NOSYS              D 0.140   rho to rep +0.995
block  50  rep msvtx_nRPC  (D 0.134)
            msvtx_nRPCMiddleStation                 D 0.119   rho to rep +0.925
block  52  rep muonAllSegments_counts/nEOL  (D 0.116)
            nEOL_NOSYS                              D 0.107   rho to rep +0.901
block  61  rep msegUnAssoc_counts/nBML  (D 0.069)
            muonAllSegments_counts/nBML             D 0.062   rho to rep +0.987

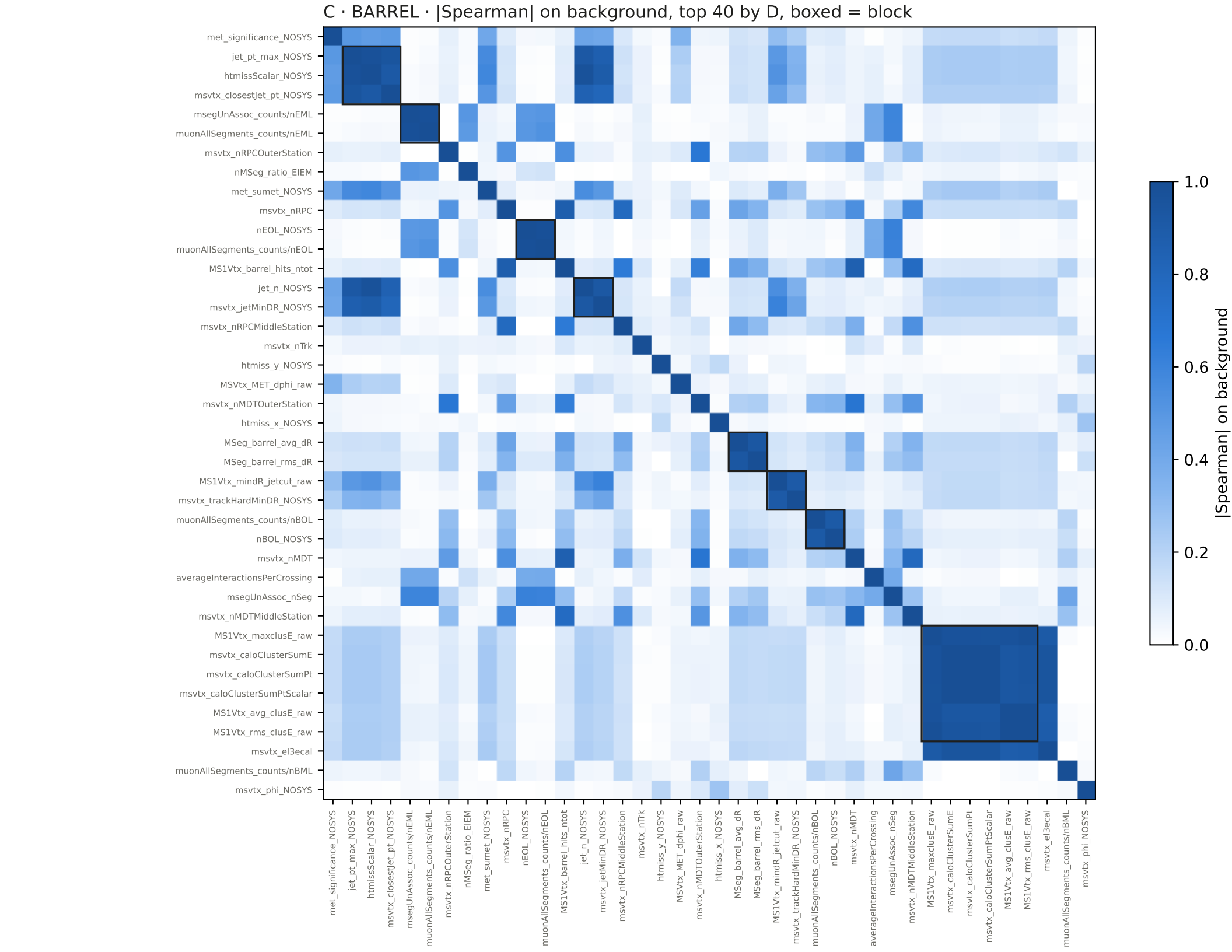
singletons: 52
```

C · ENDCAP · NN1-NN2 cross-correlation (mk4)

mk4 endcap: the 20 most correlated NN1-NN2 feature pairs on background (|Spearman|)

0.775	NN1 msvtx_caloClusterSumE	NN2 MS1Vtx_maxclustime_raw
0.743	NN1 msvtx_el3ecal	NN2 MS1Vtx_maxclustime_raw
0.731	NN1 MS1Vtx_maxclusE_raw	NN2 MS1Vtx_maxclustime_raw
0.727	NN1 msvtx_el2ecal	NN2 MS1Vtx_maxclustime_raw
0.723	NN1 msvtx_caloClusterSumE	NN2 MS1Vtx_rms_clustime_raw
0.702	NN1 MS1Vtx_maxclusE_raw	NN2 MS1Vtx_rms_clustime_raw
0.688	NN1 msvtx_el3ecal	NN2 MS1Vtx_rms_clustime_raw
0.684	NN1 MS1Vtx_rms_clusE_raw	NN2 MS1Vtx_maxclustime_raw
0.675	NN1 MS1Vtx_rms_clusE_raw	NN2 MS1Vtx_rms_clustime_raw
0.671	NN1 msvtx_el2ecal	NN2 MS1Vtx_rms_clustime_raw
0.659	NN1 MS1Vtx_avg_clusE_raw	NN2 MS1Vtx_rms_clustime_raw
0.659	NN1 MS1Vtx_avg_clusE_raw	NN2 MS1Vtx_maxclustime_raw
0.611	NN1 msvtx_el4ecal	NN2 MS1Vtx_maxclustime_raw
0.575	NN1 msvtx_el4ecal	NN2 MS1Vtx_rms_clustime_raw
0.574	NN1 msvtx_el1ecal	NN2 MS1Vtx_maxclustime_raw
0.526	NN1 msvtx_el1ecal	NN2 MS1Vtx_rms_clustime_raw
0.483	NN1 msvtx_el1hcal	NN2 MS1Vtx_maxclustime_raw
0.478	NN1 msvtx_el2hcal	NN2 MS1Vtx_maxclustime_raw
0.463	NN1 msvtx_el1hcal	NN2 MS1Vtx_rms_clustime_raw
0.452	NN1 msvtx_el2hcal	NN2 MS1Vtx_rms_clustime_raw

For ABCD the two networks must be independent on background; high pairs here are where NN1 and NN2 share information.


C · BARREL · |Spearman| on background, top 40 by D, boxed = block
met_significance_NOSYS
jet_pt_max_NOSYS
htmissScalar_NOSYS
msvtx_closestJet_pt_NOSYS
msegUnAssoc_counts/nEML
muonAllSegments_counts/nEML
msvtx_nRPCOuterStation
nMSeg_ratio_EIEM
met_sumet_NOSYS
msvtx_nRPC
nEOL_NOSYS
muonAllSegments_counts/nEOL
MS1Vtx_barrel_hits_ntot
jet_n_NOSYS
msvtx_jetMinDR_NOSYS
msvtx_nRPCMiddleStation
msvtx_nTrk
htmiss_y_NOSYS
MSVtx_MET_dphi_raw
msvtx_nMDTOuterStation
htmiss_x_NOSYS
MSeg_barrel_avg_dR
MSeg_barrel_rms_dR
MS1Vtx_mindR_jetcut_raw
msvtx_trackHardMinDR_NOSYS
muonAllSegments_counts/nBOL
nBOL_NOSYS
msvtx_nMDT
averageInteractionsPerCrossing
msegUnAssoc_nSeg
msvtx_nMDTMiddleStation
MS1Vtx_maxclusE_raw
msvtx_caloClusterSumE
msvtx_caloClusterSumPt
msvtx_caloClusterSumPtScalar
MS1Vtx_avg_clusE_raw
MS1Vtx_rms_clusE_raw
msvtx_el3ecal
muonAllSegments_counts/nBML
msvtx_phi_NOSYS
|Spearman| on background
1.0
0.8
0.6
0.4
0.2
0.0

C · BARREL · redundancy blocks

```
69 blocks; a feature joins a block only if |rho| >= 0.9 with every member;
the first member (strongest separator) is the representative.

block   2  rep jet_pt_max_NOSYS  (D 0.873)
            htmissScalar_NOSYS                      D 0.798   rho to rep +0.975
            msvtx_closestJet_pt_NOSYS               D 0.707   rho to rep +0.949
block   3  rep msegUnAssoc_counts/nEML  (D 0.858)
            muonAllSegments_counts/nEML             D 0.848   rho to rep +0.983
block   8  rep nEOL_NOSYS  (D 0.710)
            muonAllSegments_counts/nEOL             D 0.692   rho to rep +0.983
block  10  rep jet_n_NOSYS  (D 0.562)
            msvtx_jetMinDR_NOSYS                    D 0.555   rho to rep -0.923
block  17  rep MSeg_barrel_avg_dR  (D 0.366)
            MSeg_barrel_rms_dR                      D 0.297   rho to rep +0.926
block  18  rep MS1Vtx_mindR_jetcut_raw  (D 0.365)
            msvtx_trackHardMinDR_NOSYS              D 0.320   rho to rep +0.906
block  19  rep muonAllSegments_counts/nBOL  (D 0.333)
            nBOL_NOSYS                              D 0.319   rho to rep +0.907
block  22  rep msegUnAssoc_nSeg  (D 0.314)
            muonSegments_nSeg                       D 0.274   rho to rep +0.976
block  24  rep MS1Vtx_maxclusE_raw  (D 0.312)
            msvtx_caloClusterSumE                   D 0.310   rho to rep +0.966
            msvtx_caloClusterSumPt                  D 0.308   rho to rep +0.965
            msvtx_caloClusterSumPtScalar            D 0.307   rho to rep +0.965
            MS1Vtx_avg_clusE_raw                    D 0.304   rho to rep +0.966
            MS1Vtx_rms_clusE_raw                    D 0.299   rho to rep +0.987
block  25  rep msvtx_el3ecal  (D 0.293)
            msvtx_nCaloClusters_NOSYS               D 0.281   rho to rep +0.919
            msvtx_nCaloClusters_Egt300MeV_NOSYS     D 0.281   rho to rep +0.922
            msvtx_el2ecal                           D 0.267   rho to rep +0.936
block  26  rep muonAllSegments_counts/nBML  (D 0.292)
            msegUnAssoc_counts/nBML                 D 0.277   rho to rep +0.982
block  41  rep msegNCB_counts/nEIL  (D 0.168)
            muonAllSegments_counts/nEIL             D 0.132   rho to rep +0.935
block  47  rep msegUnAssoc_counts/nBIL  (D 0.152)
            muonAllSegments_counts/nBIL             D 0.146   rho to rep +0.952
block  53  rep msvtx_trackSoftSumPt_NOSYS  (D 0.111)
            msvtx_trackSoftSumNTracks               D 0.110   rho to rep +0.996
block  55  rep msvtx_eta_NOSYS  (D 0.104)
            msvtx_z                                 D 0.096   rho to rep +0.962
block  56  rep MSTracklet_barrel_avg_dR  (D 0.097)
            MSTracklet_barrel_rms_dR                D 0.097   rho to rep +1.000

singletons: 53
```

C · BARREL · NN1-NN2 cross-correlation (mk4)

mk4 barrel: the 20 most correlated NN1-NN2 feature pairs on background (|Spearman|)

0.849	NN1 msvtx_caloClusterSumE	NN2 MS1Vtx_maxclustime_raw
0.848	NN1 msvtx_caloClusterSumPtScalar	NN2 MS1Vtx_maxclustime_raw
0.826	NN1 msvtx_caloClusterSumE	NN2 MS1Vtx_rms_clustime_raw
0.825	NN1 msvtx_caloClusterSumPtScalar	NN2 MS1Vtx_rms_clustime_raw
0.823	NN1 msvtx_el2ecal	NN2 MS1Vtx_maxclustime_raw
0.817	NN1 msvtx_el3ecal	NN2 MS1Vtx_maxclustime_raw
0.808	NN1 MS1Vtx_maxclusE_raw	NN2 MS1Vtx_maxclustime_raw
0.807	NN1 MS1Vtx_maxclusE_raw	NN2 MS1Vtx_rms_clustime_raw
0.793	NN1 MS1Vtx_rms_clusE_raw	NN2 MS1Vtx_rms_clustime_raw
0.786	NN1 MS1Vtx_avg_clusE_raw	NN2 MS1Vtx_rms_clustime_raw
0.780	NN1 MS1Vtx_rms_clusE_raw	NN2 MS1Vtx_maxclustime_raw
0.779	NN1 msvtx_el3ecal	NN2 MS1Vtx_rms_clustime_raw
0.773	NN1 msvtx_el2ecal	NN2 MS1Vtx_rms_clustime_raw
0.768	NN1 MS1Vtx_avg_clusE_raw	NN2 MS1Vtx_maxclustime_raw
0.711	NN1 msvtx_el1ecal	NN2 MS1Vtx_maxclustime_raw
0.656	NN1 msvtx_el1ecal	NN2 MS1Vtx_rms_clustime_raw
0.599	NN1 msvtx_el2hcal	NN2 MS1Vtx_maxclustime_raw
0.571	NN1 msvtx_el2hcal	NN2 MS1Vtx_rms_clustime_raw
0.507	NN1 msvtx_el4ecal	NN2 MS1Vtx_maxclustime_raw
0.487	NN1 msvtx_el3hcal	NN2 MS1Vtx_maxclustime_raw

For ABCD the two networks must be independent on background; high pairs here are where NN1 and NN2 share information.